Started on	2024-02-29 00:58
State	Finished
Completed on	2024-02-29 03:41
Time taken	2 hours 43 mins
Grade	**15.05** out of 20.00 (**75**%)

Question **1**
Correct
Mark 1.00 out of 1.00

Let $P(x)$ be the statement "The word x contains the letter a ''

Find truth values of

a. $P(orange)$ TRUE ✓

b. $P(lemon)$ FALSE ✓

c. $P(true)$ FALSE ✓

d. $P(false)$ TRUE ✓

e. $P(Anna)$ TRUE ✓

Question **2**
Correct
Mark 2.00 out of 2.00

State the value of x after the following piece of code

$x := 3; \quad y := 4;$

If $P(x,y)$ then $x := y+1;$

in the following separated cases

NOTE: in the given boxes type only a number, for example if you think that $x=3$ type 3

a. $P(x,y) = "x+y>0"$ 5 ✓

b. $P(x,y) = "x^2+y^2=1"$ 3 ✓

Question 3
Complete
Mark 1.00 out of 1.00

For each of the statements below find a domain for which the statement is true and a domain for each the statement is false.

a. Everyone speaks Uzbek

b. There is someone older than 20 years

Type your answers in the box below.

a)
students who study at schools with Uzbek language as a primary- true
Uzbekistan - false
b)
in an alcohol store - true
first grade students - false

Comment:

Question 4
Partially correct
Mark 1.75 out of 2.00

In each case below decide if the proposition $\forall x P(x)$ true or false over all real numbers, and if it is false choose an appropriate counterexample, if it is true choose NONE, where $P(x)$ is

a. " $x+1>x$ " TRUE ✓ ✗

b. " $x+2=3$ " FALSE ✓ x=3 ✓

c. " $x^2+2x-3=0$ " FALSE ✓ x=2 ✓

d. " $x^2+2x-3>0$ " FALSE ✓ x=1 ✓

Question 5
Partially correct
Mark 1.00 out of 2.00

In each case below decide if the proposition $\exists x P(x)$ is true or false over all real numbers, and if it is true choose an appropriate witness, if it is false choose NONE, where $P(x)$ is

a. " $x+1>x$ " TRUE ✓ ✗

b. " $x+2=3$ " TRUE ✓ ✗

c. " $x^2+2x-3=0$ " TRUE ✓ ✗

d. " $x^2+x+1<0$ " FALSE ✓ x=1 ✗

Question 6

Complete

Mark 0.00 out of 2.00

Consider the predicate $P(x) = "x^3 - 5x + 4 = 0"$ and the domain $\{-1, 0, 1\}$.

Rewrite the propositions

a. $\forall x P(x)$

b. $\exists x P(x)$

using conjunctions and disjunctions and in each case decide it is true or false.

Write your solution on any paper, take a picture and attach it below.

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Comment:

Question 7

Complete

Mark 2.00 out of 2.00

Translate each of these statements into logical expression using predicates quantifiers and logical operations.

Denote $c =$ "A tool is in the correct place" and $e =$ "A tool is in excellent conditions"

The domain is all tools in your tool-box.

a. "Something is not in the correct place"

b. "All tools are in the correct place and are in excellent conditions''

c. "Nothing is in the correct place and is in excellent conditions''

d. "One of your tools is not in the correct place but it is in excellent condition''

Write your answer on any paper, take a picture and attach it below.

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Comment:

Question 8

Complete

Mark 1.00 out of 1.00

i. Write the meaning of $\exists x \forall y (xy = 0)$

ii. Is this statement true or false over all real numbers?

Type your answers in the box below.

i) for every real number y there is a real number x such that xy = 0
ii) true

Comment:

Question 9
Complete
Mark 1.00 out of 1.00

Let $F(x, y)$ be the statement " x can fool y '', where the domain consists of all people in the world.

Use quantifiers to express each of these statements.

a. "Everyone can fool somebody"'

b. "Everyone can be fooled by somebody''

Write your answer on any paper, take a picture and attach it below.

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Comment:

Question 10
Complete
Mark 0.50 out of 1.00

Translate the following nested quantification into an English statement that expresses a mathematical fact:

$\exists x \exists y((x \leq 0) \wedge (y \leq 0) \wedge (x - y > 0))$

Type your answer in the box below.

there is a real number x, and for this x there is a real number y, together they satisfy the expression.

Comment:

Question 11
Complete
Mark 2.00 out of 2.00

Let $C(x, y)$ mean that student x is enrolled in class y, where the domain for x consists of all students in the university and the domain for y consists of all classes here. Express the following statement by a SIMPLE English sentence:

$\exists x \exists y \forall z((x \neq y) \wedge (C(x, z) \rightarrow C(y, z)))$

There exist two students such that for any class, if one student is enrolled in it, then the other student is also enrolled in it, but the two students are not the same.

Comment:

Question 12
Complete
Mark 1.00 out of 2.00

Let $I(x)$ be the statement "Student x has an Internet connection'' and $C(x, y)$ be the statement "Students x and y have chatted over the Internet''.

The domain for variables x and y consists of all students in your class.

Use quantifiers to express the statement:

"Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class''

Write your answer on any paper, take a picture and attach it below.

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Comment:

Question 13
Complete
Mark 0.80 out of 1.00

Rewrite the statement below so that negations appear only within predicates:

$\neg\exists y(Q(y) \wedge \forall x \neg R(x, y))$

Attach your solution below.

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Comment: